

ARJUN L

Software Development Engineer

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SUMMARY

Backend-leaning full-stack engineer who likes shipping things people actually use, not just prototypes. Co-founded Lyptron, a small product studio, building an e-commerce site with a WhatsApp checkout flow, a construction app now used daily by 30+ workers, and a portfolio site for a client, among others. Outside client work, spends a lot of time on multi-agent AI systems built with LangChain, going past basic single-pass RAG into supervisor and audit-style agent setups. Currently looking for an SDE-1 role in backend or full-stack engineering.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Dart

Frontend: React, Flutter, Electron

Backend: Node.js, Express.js, REST APIs, Firebase

Databases: MongoDB, Firestore, PostgreSQL (pgvector), Pinecone

AI: LangChain, Multi-Agent Systems, RAG, Embeddings, Vector Search, Prompt Engineering

Tools: Git, GitHub, Docker, Postman, VS Code

PROJECTS

Construction Ops Multi-Agent System

Personal Project — Multi-Agent Architecture and Backend

TypeScript, Node.js, Express.js, React, PostgreSQL (pgvector), Claude, OpenAI Embeddings, Docker

- Architected a supervisor-based multi-agent RAG system for construction operations, replacing a single vector-search pipeline with three domain-specialist agents (Payment, Contract, Progress) that each retrieve through typed, access-controlled tools instead of open-ended prompts.
- Built a Supervisor agent that decomposes questions into per-specialist sub-tasks and merges their evidence, plus an Audit agent that verifies every factual claim against retrieved evidence and triggers a targeted retry when a claim is unsupported or contradicted.
- Enforced zone-based access control at the tool layer via parameterized SQL (not prompt instructions), and instrumented full request tracing (agent calls, tool I/O, latencies) to PostgreSQL, streamed live to a React UI over Server-Sent Events.
- Built a reproducible evaluation harness benchmarking the multi-agent system against plain-RAG and single-agent baselines on citation correctness and unsupported-claim rate; containerized the full stack with Docker Compose.

Nirman

Client Project (via Lyptron) — Backend Lead

Flutter, Provider, Firebase Auth, Razorpay

- Solved a real tracking gap for construction teams by building the backend for a 15-screen Flutter app that ties work-progress logs directly to payment status. Now used daily by 30+ contractors and workers across 3 client construction firms.
- Implemented state management using Provider to keep work-progress and payment data synchronized in real time across all 15 screens, with project-code-based access and role-based permissions.
- Integrated Firebase Authentication for secure contractor/worker login, and the Razorpay payment gateway so every payment is linked to its corresponding work-completion record.

Idyani

Client Project (via Lyptron) — Backend Lead

Electron, Node.js, Firebase

- Built a backend that converts Western piano input into Carnatic swara notation in real time, closing a practice-feedback gap for piano-trained students learning Carnatic music; supports 50+ students, 5 teachers, and 3 composers.
- Developed a note-evaluation system that scores timing accuracy and note placement in real time, giving teachers structured, note-by-note feedback instead of a single pass/fail grade.
- Implemented Firebase Authentication and session handling for student-teacher bookings and payments; the teacher dashboard tracks individual progress across all students.

EXPERIENCE

Co-Founder & Lead Developer — Lyptron

Web, SaaS, Mobile and AI Product Studio | Apr 2026 – Present

- Co-founded Lyptron, a product studio delivering websites, SaaS, mobile apps, and AI automation for founders, and lead all technical development as the sole developer.
- Delivered Vriddhi Vastra (Next.js, MongoDB), a full e-commerce site for a silk-saree retailer, built end-to-end with a WhatsApp-based order flow in place of a payment gateway.
- Built MadhuPortfolio, a personal portfolio website for a client, live at nmadhukumar.com, with a custom, responsive front end.
- Delivered Nirman and Idyani, which together now serve 85+ users across construction and music education (detailed under Projects).

Customer Experience — Supertails

Bangalore, India | Oct 2025 – Present

- Supported Supertails' customer-support chatbot by tagging real customer queries with the correct intent, reviewing responses for accuracy and tone, and flagging incorrect ones before they reached more customers.
- Built a standalone tool (React frontend, Flask backend calling a third-party AI-image-detection API) that lets agents check whether a customer-submitted refund/damage claim image is AI-generated before approving the claim.
- Helped onboard other agents onto the claim-image tool when it launched.

EDUCATION

B.Tech in Mathematics and Computing — M S Ramaiah University of Applied Sciences

Bangalore, India | 2021 – 2025

Relevant Coursework: Statistical Methods, Linear Algebra, Database Systems, Probability Theory, Operations Research